



Rise of the apes

Africa's endangered Mountain Gorillas, earlier on the verge of extinction, is the only great ape that seems to be growing <http://dgit.in/ApeRise>



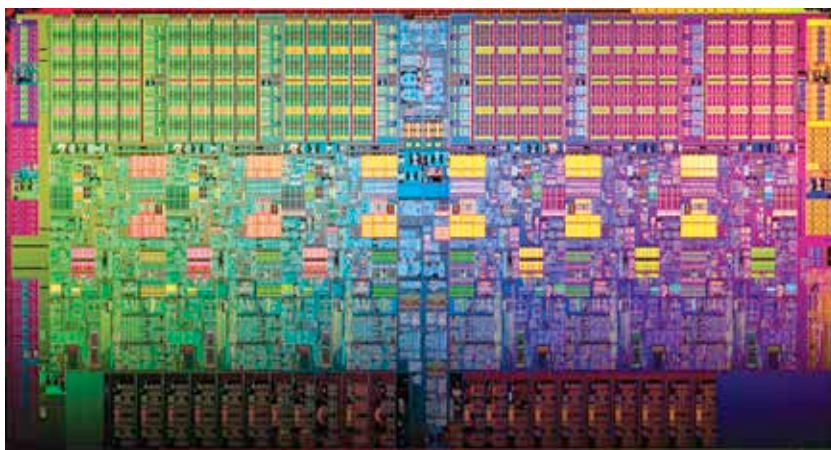
Vacation scammer to be fined

The FCC has proposed to a fine \$120M for the vacation scammer who made 96 million phone calls over 3 months. <http://dgit.in/VacSpam>

Intel's new Mesh Architecture

The race to pump more cores into your processor has begun heating up and the Mesh Architecture is Intel's way of building scalability.

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icroprocessor cores have only been increasing over the years. It's one of the key methods by

which the processor compute power is scaled. You can increase the core clock speed and/or you can increase the core count and get more processing power. However, these two methods are easier said than done, as each method has its own set of implications which can be either negative or positive. It's akin to receiving medication – you get the desired effect but there might be a few side-effects thrown in the mix as well.

RING BUS ARCHITECTURE – THE CURRENT SOLUTION

When it comes to increasing clock speed, the obvious implication is the increased power consumption and the resultant extra heat. And when you think of increasing the number of cores, you need to wonder about optimising the time taken for one core to talk to another. Think of it like a room in which you let more people in, you can either have everyone talking at the exact same time resulting in

cacophony or you can have a system of communication wherein a person raises their hand to be heard. Processors face a similar dilemma.

Back in the simple days, each core within the CPU had its own dedicated or private interconnect to the L3 cache or the Last Level Cache. Physically, this meant that there would be hundreds (if not thousands) of tiny wires between each core and the L3. As more cores were needed, this meant that either an additional set of thousands of

wires needed to be laid down or a new method of communication had to be introduced. This is when the Ring Bus Architecture came into being.

Cores were placed adjacent to each other in uniform stacks and the interconnect would pass through one stack before hitting the L3 cache. While this did introduce a slight amount of latency between certain cores and the L3 cache, the net result was a much more optimised manner of interconnect. This method of arranging cores in a ring wasn't new. The world of networking had been using ring networks for ages. However, each system comes with its own disadvantages.

Like the previous method, wherein each core had its own interconnect path to the L3 cache, had a limitation of the number of cores it could optimally handle, even the Ring Bus architecture had a limit. The issue here was again latency while communicating between cores. With Skylake-SP and Skylake-X processors, Intel is introducing a new architecture - Intel Mesh Architecture. Again, if we are to look back at networking, you'll learn that mesh networks have been in existence for

